

CHE 311 Homework Week 2

1. Velocity profiles and the stress components

For each of the following velocity distributions, find all the components of $\nabla \mathbf{v}$ and $\rho \mathbf{v} \mathbf{v}$ for the Newtonian fluid. The parameter b is a constant.

- (a) $v_x = by$, $v_y = 0$, $v_z = 0$
- (b) $v_x = by$, $v_y = bx$, $v_z = 0$
- (c) $v_x = -by$, $v_y = bx$, $v_z = 0$
- (d) $v_x = -0.5bx$, $v_y = -0.5by$, $v_z = bz$

2. Calculate the force

There is gear oil fill between the two cylinders, with viscosity $\mu = 30 \text{ Pa}\cdot\text{s}$. The inner cylinder is stationary. The outer cylinder has a constant angular rotational speed of 5 degree/second . The length of the cylinders is 1 m . The radius of the outer cylinder $R = 1 \text{ m}$. The gap distance between the two cylinders $d = 5 \text{ mm}$. Please calculate the torque applied on the outer cylinder to maintain the constant motion. (Note that the gap between the cylinders is much smaller than the radius of the cylinders.)

